This document describes three different data tables with star and gas data from a particular starforge simulation.

accreted_gas_properties.pq summary.

Each row shows the physical attributes of gas that will accrete onto a sink (star) particle at a particular moment in time. Gas that has already accreted is not included.

Table Overview

- **Index Key:** `sink_id`
- **Index Key** `snap`
- **Spatial information:** Center of mass positions (x, y, z) and velocities (u, v, w). Specific angular momentum components. Velocity dispesion. Principal axes.
- **Average gas properties:** Mass-weighted averages, sample standard deviations (`_std`), and distribution percentiles (16th, 50th/median, and 84th) for different gas properties (temperature, density, magnetic field, etc.)

Identification & Bulk Properties

Column Name	Data Type	Physical Units	Field Description
`sink_id`	`int64`	Dimensionless	Unique identifier of the sink particle (representing a star) within the simulation.
`snap`	`int64`	Dimensionless	Snapshot number. Shapshots are spaced by 2.4703e4 years.
`M_tot`	`float64`	code_mass ($M_\odot$)	Total mass of the unique, non-feedback gas particles that will accreted by the sink.

Column Name	Data Type	Physical Units	Field Description
`R_eff`	`float64`	code_length (pc)	Effective Radius (R_{eff}). This is the radius of the sphere with the same volume as the gas: $R_{\text{eff}} = \left(\frac{3 \cdot \text{Vol}}{4\pi}\right)^{1/3}$, where $\text{Vol} = \Sigma \frac{m}{\rho}$. m and ρ are the mass density of each gas particle.

Center of Mass & Kinematics

Column Name	Data Type	Physical Units	Field Description
`x_cm_1`	`float64`	code_length (pc)	x coordinate of the gas center of mass.
`x_cm_2`	`float64`	code_length (pc)	y coordinate of the gas center of mass.
`x_cm_3`	`float64`	code_length (pc)	z coordinate of the gas center of mass.
`v_cm_1`	`float64`	code_velocity (m/s)	x velocity component (u) of the gas center of mass.
`v_cm_2`	`float64`	code_velocity (m/s)	y velocity component (v) of the gas center of mass.
`v_cm_3`	`float64`	code_velocity (m/s)	z velocity component (w) of the gas center of mass.
`L_vec_1`	`float64`	code units	x component of the angular momentum vector (with respect to gas center of mass)

Column Name	Data Type	Physical Units	Field Description
L_vec_2	float64	code units	y component of the angular momentum vector
L_vec_3	float64	code units	z component of the angular momentum vector
sig_3D	float64	code_velocity (m/s)	3D Velocity Dispersion.

Spatial Morphology

Column Name	Data Type	Physical Units	Field Description
R_p_1	float64	code_length (pc)	1st principal component length
R_p_2	float64	code_length (pc)	2nd principal component length
R_p_3	float64	code_length (pc)	3rd principal component length

Thermodynamics, magnetic field, density

All statistics are mass weighted

Column Name	Data Type	Physical Units	Field Description
T	float64	Kelvin (K)	average gas temperature.
T_std	float64	Kelvin (K)	standard deviation of the gas temperature.
T_16	float64	Kelvin (K)	16th percentile of gas temperature distribution.
T_med	float64	Kelvin (K)	50th percentile (median) of gas temperature distribution.
T_84	float64	Kelvin (K)	84th percentile of gas temperature distribution.

Column Name	Data Type	Physical Units	Field Description
B	float64	Gauss (G)	average magnetic field magnitude ($B_{\mathrm{mag}} = \sqrt{B_x^2 + B_y^2 + B_z^2}$).
B_std	float64	Gauss (G)	standard deviation of the magnetic field magnitude.
B_16	float64	Gauss (G)	16th percentile of magnetic field magnitude.
B_med	float64	Gauss (G)	50th percentile (median) of magnetic field magnitude.
B_84	float64	Gauss (G)	84th percentile of magnetic field magnitude.
Ne	float64	Dimensionless	average electron abundance (ratio of free electrons per baryon).
Ne_std	float64	Dimensionless	standard deviation of electron abundance.
Ne_16	float64	Dimensionless	16th percentile of electron abundance.
Ne_med	float64	Dimensionless	50th percentile (median) of electron abundance.
Ne_84	float64	Dimensionless	84th percentile of electron abundance.
rho	float64	code_density	average gas volume density (ρ).
rho_std	float64	code_density	standard deviation of gas density.

Column Name	Data Type	Physical Units	Field Description
`rho_16`	`float64`	code_density	16th percentile of gas density.
`rho_med`	`float64`	code_density	50th percentile (median) of gas density.
`rho_84`	`float64`	code_density	84th percentile of gas density.

Energetics

Column Name	Data Type	Physical Units	Field Description
`E_grav`	`float64`	code_units	Gravitational Potential Energy ($E_{\rm pot}$). Evaluated with `pytreegrav.Potential` using particle masses, positions, and smoothing lengths (h).
`E_kin`	`float64`	code_units	Kinetic Energy ($E_{\rm kin}$) in the center of mass frame of the gas particles.
`E_mag`	`float64`	code_units	Magnetic Energy ($E_{\rm mag}$). Integrated over local gas cell volumes: $E_{\rm mag} = \frac{1}{8\pi}\sum B^2 V$.
`E_int`	`float64`	code_units	Internal Thermal Energy ($E_{\rm int}$). A summation of gas masses multiplied by their specific internal energies.

Particle Tracking Counters

Column Name	Data Type	Physical Units	Field Description
N_inc	int64	Count	Number of unique, non-feedback gas particles that will accrete onto the target sink.
N_fb	int64	Count	Number of feedback gas cells encountered and excluded from analysis.

accreted_gas_properties_pre_stellar.pq summary

Same as accreted_gas_properties.pq, except that there is only one row per star, corresponding to the snapshot before that star appeared.

first_snap_table.pq

Table containing initial positions, final masses, and initial masses for all the stars

Data

Column Name	Data Type	Physical Units	Field Description
pid	int64	Dimensionless	star id
snap	int64	Dimensionless	
x	float64	code_length (pc)	x coordinate of star at initial snapshot.
y	float64	code_length (pc)	y coordinate of star at initial snapshot.
z	float64	code_length (pc)	z coordinate of star at initial snapshot.

Column Name	Data Type	Physical Units	Field Description
`mf`	`float64`	code_mass ($M_\odot$)	Final mass of star (at end of simulation or at star death–whichever comes first).
`mstar`	`float64`	code_mass ($M_\odot$)	Mass of the star at the initial snapshot